CAPSTONE PROJECT

Multimodal Brain Tumor Classification Using BLIP and CLIP

A Comparative Study on Accuracy and Performance

Name :Nithin Nayak(2301ai16)

Department of Artificial Intelligence & Data Science
Professer :-Jimson Mathew

97.94%

CLIP Test Accuracy

96.49%

BLIP Test Accuracy

7,023

MRI Images

4

Tumor Classes

From CNNs to Vision-Language Models in Medical Imaging

- Traditional CNNs (ResNet, VGG, DenseNet) dominate medical image classification but suffer from limited generalization, poor domain adaptability, and brittleness under distribution shift in real-world MRI conditions.
- Vision-Language Models (VLMs) — CLIP (Contrastive Language-Image Pretraining) and BLIP (Bootstrapped Language-Image Pretraining) — represent a paradigm shift: leveraging large-scale image-text pretraining to produce semantically rich embeddings via Vision Transformer (ViT) encoders.
- While VLMs show strong zero-shot and transfer learning capability, their applicability in domain-specific medical imaging remains underexplored — creating the central research gap this study addresses.
- This study proposes supervised fine-tuning of both BLIP and CLIP on a curated 7,023-image MRI dataset, moving beyond zero-shot paradigms to maximise domain-specific diagnostic performance.

Limitations of Zero-Shot VLM Classification in Medical Imaging

- Zero-shot VLMs rely on general-purpose embeddings trained on Internet-scale image-text data — these fail to capture fine-grained pathological distinctions critical in MRI classification (e.g., glioma vs. meningioma boundary textures).
- Domain shift from natural images to clinical MRI scans introduces embedding misalignment, where class-label prompts like 'a photo of a glioma' carry insufficient anatomical specificity.
- Class imbalance and visual similarity between tumour types (particularly glioma and meningioma) amplify false negative risk — a clinically unacceptable outcome in oncological diagnosis.
- This motivates supervised fine-tuning of vision encoders with frozen text components, ensuring domain-specific visual representations while retaining pre-trained semantic structure.

Dataset: 7,023 Brain MRI Images Across Four Classes

Tumour Class	Train Images	Test Images	Total	Source Dataset
Glioma	1,321	300	1,621	Kaggle + Figshare
Meningioma	1,339	306	1,645	Kaggle + Figshare
No Tumor	1,595	405	2,000	Br35H
Pituitary	1,457	300	1,757	Kaggle + Figshare
TOTAL	**5,712**	**1,311**	**7,023**	**3 Public Repos**

80/20 stratified train/test split (seed=42) · 5-fold CV · ~52% male · ages 18–75 · T1-weighted axial de-identified MRIs

Robust Preprocessing for Domain Adaptability

Resize

224×224 px (CLIP/BLIP input); constant input dimensionality across all sources.

Normalize

Pixel values scaled to [–1, 1] (mean=0.5, std=0.5) for gradient stability.

Augmentation

Random horizontal/vertical flips (train-only); zoom, rotation to reduce overfitting.

Contrast & Brightness

Variation simulation to account for cross-scanner MRI acquisition differences.

Gaussian Noise

Injected noise ($\mu=0$, $\sigma^2=0.02$) simulates MRI acquisition noise patterns.

FGSM Adversarial

$\varepsilon=0.01$ perturbation along gradient direction for adversarial robustness testing.

PGD Attack Simulation

7-step iterative perturbation (step size 0.01) to model strong adversarial threats.

Text Embedding

Class-label prompts frozen; only vision encoder last-2 layers fine-tuned.

Dual-Encoder Supervised Fine-Tuning Paradigm

MRI Input
7,023 Axial T1 Scans (4 classes)

→ **Preprocessing**
Resize · Norm Augmentation

→ **Vision Encoder**
ViT-based (Last 2 layers unfrozen)

→ **Text Encoder**
Class-label prompts (FROZEN)

→ **Fusion Block**
Cosine Sim / Cross-Attention

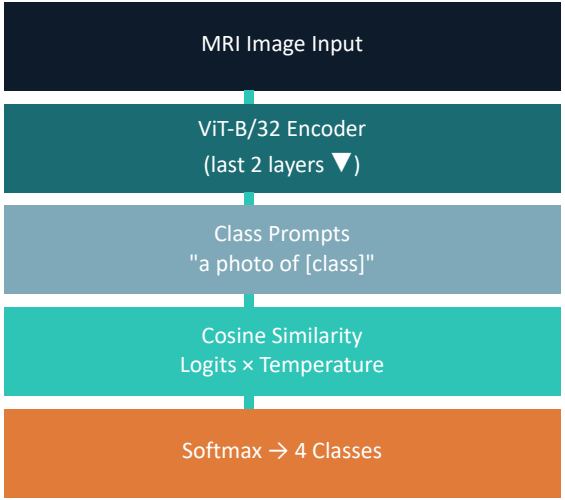
→ **Classification Head**
768→512→4 BatchNorm+ReLU

→ **Prediction**
Glioma / Meningioma Pituitary / None

- Both CLIP and BLIP share the same supervised fine-tuning paradigm: vision encoder last-2 layers unfrozen, text encoder fully frozen, classification head trained end-to-end.
- Dual-encoder fusion aligns visual embeddings with class-label text embeddings in a shared latent space using cosine similarity (CLIP) or cross-attention (BLIP).
- Contrastive loss pulls similar image-text pairs together while pushing dissimilar pairs apart, reinforcing tumour-specific semantic associations.

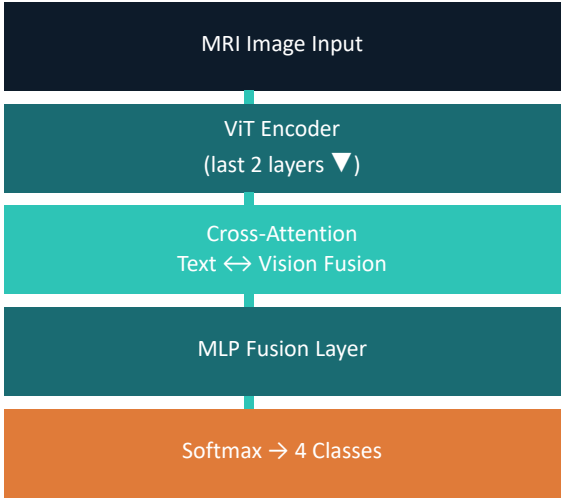
CLIP: Contrastive Language-Image Pretraining

- Pre-training Objective: CLIP is trained via contrastive loss on 400M image-text pairs, learning a joint embedding space where semantically aligned image-text pairs have high cosine similarity.
- Vision Encoder: ViT-B/32 backbone extracts CLS token embeddings (dim=768) capturing global and hierarchical local semantics from MRI patches.
- Fine-Tuning Strategy: Text encoder frozen; last 2 layers of vision encoder unfrozen + a linear head (768→4). Text embeddings pre-computed from class-label prompts and normalised.
- Fusion Mechanism: Forward pass computes cosine similarity between vision features and fixed text embeddings → scaled logits → softmax classification.



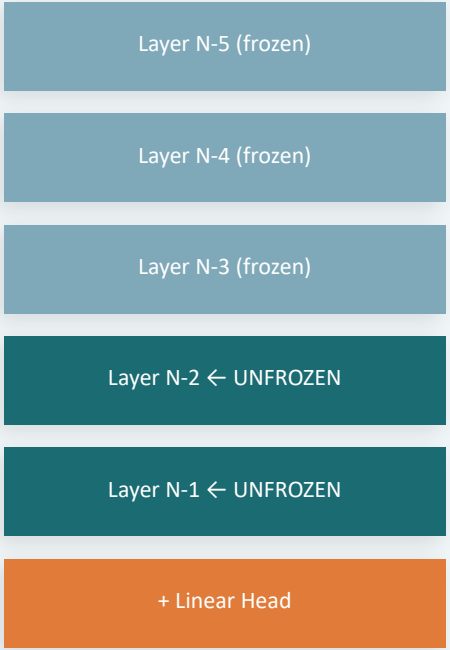
BLIP: Bootstrapped Language-Image Pretraining

- BLIP extends contrastive learning with a bootstrapped pre-training pipeline: a captioner generates synthetic captions and a filter removes noisy ones, enabling richer multimodal representations.
- Cross-Attention Mechanism: Text embeddings modulate visual features through self-attention layers in a joint encoder-decoder framework, providing finer semantic alignment between image regions and textual concepts.
- Fine-Tuning Strategy: Majority of layers frozen to preserve general knowledge; final 2 vision encoder layers selectively unfrozen for MRI adaptation. Custom classification head appended.
- Key Advantage: BLIP's cross-attention enables it to attend to localised tumour features (boundaries, textures) rather than global image statistics — yielding more balanced precision and recall across all four classes.



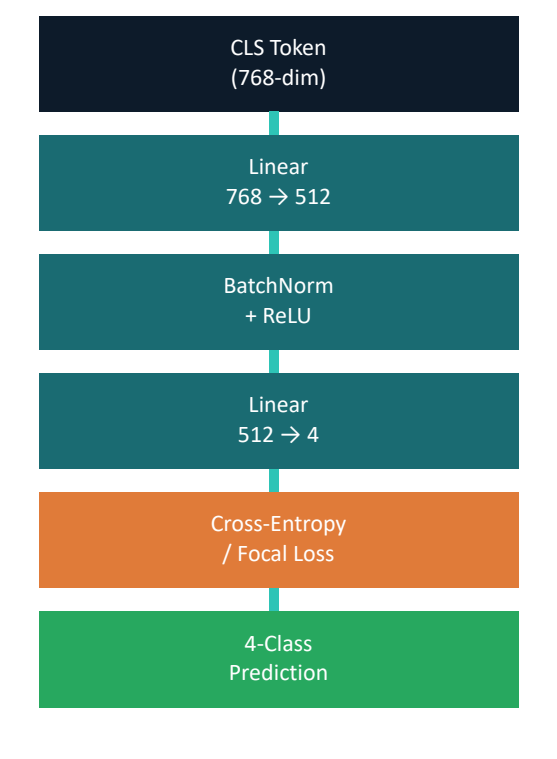
Selective Layer Unfreezing: Preserving Pretrained Knowledge

- Frozen Backbone: The bulk of both CLIP and BLIP vision encoders remain frozen, preserving general-purpose hierarchical features learned from large-scale pretraining (e.g., edge detection, texture, shape) — reducing overfitting risk on the 7,023-image medical dataset.
- Unfrozen Last 2 Vision Layers: Only the final two transformer blocks of the ViT are unfrozen. These layers are responsible for task-specific feature aggregation — unfreezing them allows adaptation to MRI-specific patterns such as tumour morphology and tissue heterogeneity.
- Text Encoder Fully Frozen: Text embeddings for class-label prompts are pre-computed and fixed throughout training. This prevents semantic drift and ensures consistent alignment between visual features and class concepts.
- Rationale: This asymmetric fine-tuning strategy balances computational efficiency (fewer trainable parameters) with domain adaptation — avoiding catastrophic forgetting while enabling MRI-specific visual learning.



Custom Head: Linear 768 → 512 → 4 with BatchNorm & ReLU

- Layer 1 — Linear (768→512): Reduces ViT CLS token embedding dimensionality, learning a compressed tumour-discriminative feature representation from the high-dimensional visual space.
- BatchNorm + ReLU: Batch normalisation stabilises gradient flow by normalising activations per mini-batch; ReLU introduces non-linearity and enforces sparse activation — critical for distinguishing visually similar tumour sub-types.
- Layer 2 — Linear (512→4): Final projection to four logits corresponding to glioma, meningioma, no-tumor, and pituitary — activations passed through cross-entropy loss during training.
- Loss Function: Hybrid of focal loss and cross-entropy addresses class imbalance. Focal loss down-weights easy negatives, focusing training on hard boundary cases (glioma vs. meningioma) — reducing false negative rate in minority tumour classes.



Hyperparameters: AdamW + Cosine Annealing, 30 Epochs

Optimiser: AdamW

Learning rate 3×10^{-4} with default weight decay. AdamW decouples weight decay from gradient updates, providing stronger regularisation than Adam — critical for preventing overfitting on the ~5,700 training images.

Scheduler: Cosine Annealing

Cosine Annealing Warm Restarts (T_0=5): learning rate follows a cosine curve, resetting every 5 epochs. This escapes local minima and promotes convergence to wider, more generalisable optima.

Batch Size: 16 · Epochs: 30

Batch size of 16 balances GPU memory constraints (CPU-only: Intel i5-9300H, 8GB RAM) with stable gradient estimates. 30 epochs ensured convergence without excessive training time (~4–6 hours/epoch on CPU).

Loss: Cross-Entropy + Focal

Cross-entropy provides multi-class probability supervision. Focal loss (γ>0) re-weights hard examples, reducing gradient contribution from correctly-classified easy samples — directly addressing class imbalance.

Hardware: Intel Core i5-9300H · CPU-only (no discrete GPU) · PyTorch 2.1 · 5-fold CV seeds: 42, 123, 456, 789, 1011

CLIP: 97.94% Test Accuracy with 22 Misclassifications

97.94%
Test Accuracy

99.40%
Train Accuracy

22
Misclassifications

0.98
Macro F1-Score

Class	Precision	Recall	F1-Score
Glioma	1.00	0.92	0.96
Meningioma	0.93	0.99	0.96
No Tumor	1.00	1.00	1.00
Pituitary	0.99	1.00	0.99
Overall / Macro	**0.98**	**0.98**	**0.98**

⚠ Key weakness: Glioma recall = 0.92 (lowest class) — primary confusion: Glioma ↔ Meningioma (22/22 errors concentrated here)

BLIP: 96.49% Accuracy with Superior Precision-Recall Balance

96.49%
Test Accuracy

97.81%
Train Accuracy

31
Misclassifications

0.96
Macro F1-Score

Class	Precision	Recall	F1-Score
Glioma	0.99	0.89	0.94
Meningioma	0.90	0.96	0.93
No Tumor	0.99	1.00	1.00
Pituitary	0.98	0.99	0.99
Overall / Macro	**0.96/0.97**	**0.96**	**0.96**

✓ BLIP advantage: More balanced precision-recall profile across all tumour classes. Cross-attention enables finer-grained discrimination of morphological tumour features, reducing systemic bias toward easy (no-tumor) cases.

COMPARATIVE ANALYSIS

CLIP vs BLIP vs CNN Baselines: Accuracy–Reliability Trade-off

Metric	CLIP	BLIP
Test Accuracy	97.94% ± 0.32%	96.49% ± 0.41%
Training Accuracy	99.40%	97.81%
Macro F1-Score	0.98	0.96
Glioma Recall	0.92	0.89
Meningioma Recall	0.99	0.96
Misclassifications	22 / 1311	31 / 1311
Class Balance	Slight bias	More balanced
FN Risk (clinical)	Moderate	Lower

McNemar's test ($p < 0.05$) confirms statistically significant differences between CLIP and BLIP predictions. Xception (literature: 99.72%) outperforms both VLMs in raw accuracy — pointing to CNN spatial feature extraction advantages in purely visual tasks.

PyTorch Implementation: Theory → Practice Alignment

Vision Encoder Fine-Tuning (CLIP)

```
# Freeze all layers then selectively unfreeze
for param in clip_model.parameters(): param.requires_grad = False
for param in
clip_model.visual.transformer.resblocks[-2:].parameters():
    param.requires_grad = True  # Last 2 layers unfrozen
classifier = nn.Sequential(
    nn.Linear(768, 512), nn.BatchNorm1d(512), nn.ReLU(),
    nn.Linear(512, 4)   # 4 tumour classes
)
```

Training Loop Core

```
optimizer = AdamW(trainable_params, lr=3e-4)
scheduler = CosineAnnealingWarmRestarts(optimizer, T_0=5)
for epoch in range(30):
    for imgs, labels in train_loader:
        vision_feats = model.encode_image(imgs)  # CLS token
        logits = classifier(vision_feats)
        loss = focal_cross_entropy(logits, labels)
        loss.backward(); optimizer.step(); scheduler.step()
```

- Data Pipeline: ImageFolder → DataLoader (batch=16) with on-the-fly augmentation perfectly mirrors the paper's preprocessing specification — flip, resize 224×224, normalise (μ=0.5, σ=0.5).
- BLIP Implementation: Analogous structure using Salesforce's BLIPModel/BLIPProcessor with cross-attention fusion replacing CLIP's cosine similarity fusion block.
- Evaluation: Classification report + Seaborn confusion matrix heatmap at each epoch; best model checkpoint saved by validation accuracy — directly reproducible from the methodology section.

Why CNNs Still Lead — And the Path to Hybrid VLMs

CURRENT LIMITATIONS

- CNN Spatial Superiority: Xception (99.72%) outperforms both VLMs using depthwise separable convolutions optimised for local spatial feature extraction — a domain where purely attention-based ViTs are structurally disadvantaged.
- Joint Embedding Noise: VLMs add noise in purely visual tasks through image-text joint embedding — cross-modal attention can suppress spatially localised tumour features in favour of global semantic alignment.
- Hardware Constraints: CPU-only training (i5-9300H) limits batch size and epoch throughput — GPU-accelerated experiments would enable larger models and deeper fine-tuning.
- No Explainability: Grad-CAM / attention heatmaps were not implemented — clinical adoption requires spatial interpretability showing which tumour regions drive classification decisions.

FUTURE DIRECTIONS

- Hybrid CNN-VLM Architectures: Combine CNN-based local spatial feature extraction (Xception backbone) with transformer-based contextual reasoning — targeting >99% accuracy with multimodal interpretability.
- Domain-Specific Pretraining: Fine-tune on large-scale medical image corpora (TCGA, ADNI, RSNA) using medical image-report pairs — reducing the domain shift that limits VLM performance in clinical settings.
- Explainability Integration: Implement Grad-CAM + attention rollout visualisations to produce tumour region attribution maps — critical for radiologist trust and regulatory approval.
- Multi-Centre Validation: Expand to T2/FLAIR MRI types across multiple scanner manufacturers; 1-year clinical trial for radiologist concordance measurement.

Conclusion & Key Takeaways

01

VLMs Are Viable — With Supervision

Supervised fine-tuning of CLIP (97.94%) and BLIP (96.49%) delivers competitive diagnostic performance vs CNN baselines — demonstrating that domain-adapted VLMs are clinically promising tools for MRI tumour classification.

02

Accuracy vs Clinical Reliability

CLIP leads in raw accuracy; BLIP leads in balanced precision-recall and lower false negative risk. For oncological applications where missing a glioma is life-threatening, BLIP's profile is the preferred clinical choice.

03

The Path Forward: Hybrid Models

CNN spatial precision + VLM contextual reasoning + explainability (Grad-CAM) + domain-specific pretraining on medical corpora represents the most promising architecture for next-generation AI diagnostics.

Thank you